Seminar case: When a self-driving car kill – Tempe 2018 to London 2026

CS5805 Ethics and Governance of Digital Systems · Lecture 2 seminar · 60 minutes

Before the seminar: read this case and Stilgoe (2021). Skim Siegel and Pappas (2023), pp. 215–221, including Table 1.

Content note: this case describes a real fatal collision. Please be respectful of the person who died and her family in discussion.

The case

1. The collision

Shortly before 10pm on 18 March 2018, Elaine Herzberg, aged 49, was walking her bicycle across Mill Avenue in Tempe, Arizona. She was crossing a dark stretch of road, away from a pedestrian crossing. A Volvo XC90 SUV travelling at about 40 mph struck and killed her.

The Volvo belonged to Uber's Advanced Technologies Group (ATG), which was testing a self-driving system on public roads. Uber ran around 40 test cars in Tempe. The car was driving itself, with one human safety driver, Rafaela Vasquez, behind the wheel. It was the first recorded death of a pedestrian caused by a fully self-driving test vehicle.

2. What the machine saw

The US National Transportation Safety Board (NTSB) reconstructed the system's decisions second by second. The car's radar and lidar first detected Herzberg 5.6 seconds before impact, in good time to stop. The software, however, could not decide what she was. It switched her classification between a vehicle, a bicycle and an unknown object.

Each time the label changed, the system treated her as a brand-new object and discarded her movement history. So, it never predicted that her path would cross the car's lane. The system had not been designed to expect pedestrians crossing away from a crossing.

The software recognised an emergency only about 1.2 seconds before impact. Even then, it did not brake hard. Uber's engineers had built in a one-second pause before emergency braking, to cut down on false alarms and jerky, uncomfortable rides. They had also switched off the Volvo's own factory-fitted collision-avoidance system while Uber's software was driving. The design relied on the human safety driver to intervene.

3. The human in the loop

Uber had originally put two operators in each test car: one to watch the road and one to monitor the software. By late 2017 it had cut this to one. Vasquez was alone, and her job was to watch the road and take over in an emergency.

Investigators found she had been looking at a phone for roughly a third of the trip, and police concluded a TV show was streaming on it. She looked up about a second before impact. Her lawyers later argued she was checking a work messaging app. The NTSB called the wider pattern "automation complacency": people who supervise automation for long periods stop expecting it to fail.

4. The organisation

The NTSB's findings went beyond the driver. Uber ATG had no standalone safety division, no formal safety plan and no dedicated safety manager. It also did little to check whether safety drivers were following the rules, such as the ban on phone use. The Board's chair summed this up as an "inadequate safety culture" and said the failings of both the software and the driver were symptoms of it.

The NTSB also found that Herzberg's impairment from drugs and her crossing outside a crosswalk contributed to the crash.

5. The regulator

Uber had moved much of its testing to Arizona after California objected to its test cars running without a state permit. Arizona had actively courted self-driving companies with a light-touch approach. The NTSB found that Arizona's transport department had exercised insufficient oversight, both before and after the crash. Days after the collision, Arizona's governor suspended Uber's testing, and Uber halted its self-driving programme nationally.

6. Who was held responsible?

- **Uber** settled with Herzberg's family within weeks, on undisclosed terms. Prosecutors decided not to bring criminal charges against the company. Uber introduced a safety management system and said its revised software would have detected Herzberg and braked more than four seconds earlier. It sold its self-driving unit to Aurora in 2020.
- **Rafaela Vasquez** was charged with negligent homicide in 2020. In July 2023 she pleaded guilty to a reduced charge of endangerment and received three years' supervised probation. Her lawyer argued that Uber should share the blame, not least for putting her in the car alone.
- **The NTSB** named the driver's distraction as the probable cause. It issued six safety recommendations to US federal regulators, Arizona, vehicle licensing officials and Uber.

7. Eight years later: London

On 3 September 2026, Uber and the British AI company Wayve launched London's first self-driving taxi service. It began with fewer than 20 cars, bookable through the Uber app. Each car carries a trained safety driver, because approval to operate without one has not yet been granted. Waymo and Baidu are preparing to launch rival services in London.

These services sit under the **Automated Vehicles Act 2024**, which the government describes as one of the most comprehensive AV laws in the world. Its main features are:

- **A safety standard.** Authorised self-driving vehicles must be at least as safe as careful and competent human drivers.
- **Responsibility moves to the company.** When a vehicle is authorised and driving itself, an "authorised self-driving entity" (usually the developer or manufacturer) is responsible for how it drives. The person in the driving seat is no longer liable for driving offences while self-driving is engaged.
- **Licensed operators** are responsible for services that run with no one in the driving seat.
- **Marketing rules.** It is an offence to describe a vehicle as self-driving if it is not authorised as such.

In a supervised pilot like the current Uber – Wayve service, the safety driver is still legally the driver.

The detailed safety standard is still being written. The Department for Transport consulted on ten draft safety principles between 17 June and 9 September 2026. The final statement is due before Parliament in 2027, with the Act fully in force from the second half of 2027.

Seminar questions

Answer all six questions. Use the ethical perspectives from Lecture 2 and support your arguments with evidence from the case and the reading.

1. Identify **three** stakeholders affected by Uber's testing programme in Tempe. For each, explain one ethical concern the case raises for them.
2. To reduce false alarms and uncomfortable rides, Uber delayed emergency braking by one second and disabled the Volvo's own collision-avoidance system. Analyse this design decision from a **utilitarian** perspective. What was traded against what, and whose interests were counted?
3. Elaine Herzberg never agreed to take part in Uber's test programme. Using **deontological or rights-based** reasoning, including Rawls's veil of ignorance, evaluate whether testing unproven self-driving cars on public roads is morally permissible.
4. The NTSB concluded that the failures of the software and the safety driver were symptoms of Uber's safety culture. Using **virtue ethics**, discuss where moral responsibility for Herzberg's death should lie with the safety driver, Uber's engineers, Uber's leadership, or the state of Arizona.
5. Compare the **governance** of Uber's testing in Arizona in 2018 with the UK's Automated Vehicles Act 2024. To what extent would the UK framework have prevented the crash, or allocated responsibility differently?
6. Eight years after Tempe, Uber is again running self-driving cars with safety drivers on public roads, this time in London. **Recommend three governance measures**, organisational or regulatory, that should apply to the London pilot. Justify them using

an ethical stance, stating whether it is universalist or relativist, and the supporting reading. *(10 marks)*

Reading

- Stilgoe, J. (2021) How can we know a self-driving car is safe? *Ethics and Information Technology*, 23(4), 635–647.
- Siegel, J. and Pappas, G. (2023) Morals, ethics, and the technology capabilities and limitations of automated and self-driving vehicles. *AI & Society*, 38, 213–226.
- Optional: Nyholm, S. and Smids, J. (2016) The ethics of accident-algorithms for self-driving cars: an applied trolley problem? *Ethical Theory and Moral Practice*, 19(5), 1275–1289.

Case sources

- NTSB (2019) Highway Accident Report NTSB/HAR-19/03
- IEEE Spectrum: NTSB investigation into the Uber crash
- NPR (28 July 2023): backup driver pleads guilty to endangerment
- AP via WTOP (3 September 2026): robotaxi service debuts in London
- GOV.UK (17 June 2026): consultation on the draft statement of safety principles